Proximity Sensor

The proximity sensor is a device used to detect nearby objects. When the object comes into its sensing area an electrical signal gets generated and this helps to detect the object.

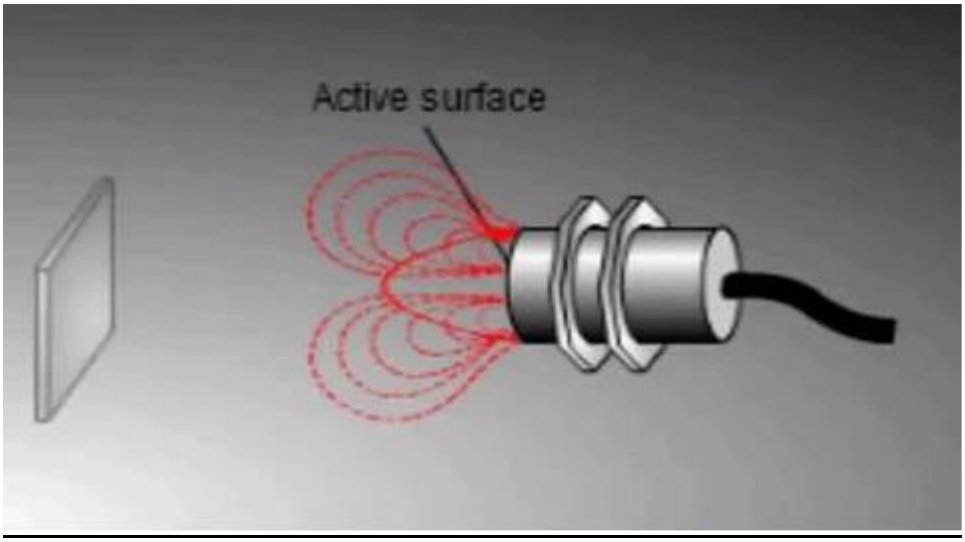
Types of Proximity Sensors and Their Uses

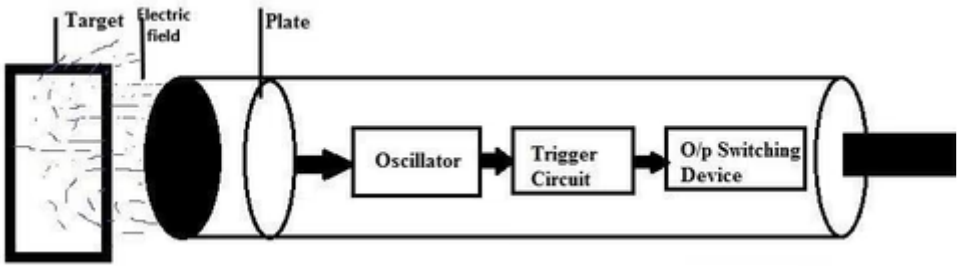
Proximity sensors are used in many applications to detect objects without physical contact. They can measure the distance, presence, speed, and direction of an object relative to the sensor itself. There are different proximity sensor types such as ultrasonic ones which emit high-frequency sound waves that bounce off a target surface; capacitive ones with two conductive plates forming a capacitor; inductive magneto-strictive ones using magnetic fields generated by current flow through coil windings or permanent magnets; photoelectric devices relying on principles involving light sources like infrared beams which activate output when penetrated by an obstacle and active optical systems utilizing laser technology for sensing objects within its line-of-sight range. For precise application requirements you must select the right type based on your needs so be sure to study each one thoroughly!

Based on the non-contact object detection method, there are five proximity sensor types. such as,

- Capacitive Proximity Sensor.
- Inductive Proximity Sensor.
- Magnetic Proximity Sensor.
- Optical Proximity Sensor.
- Ultrasonic proximity Sensor.

1. **Capacitive Proximity Sensor** capacitive proximity sensor types work on the principle of varying capacitance. So, as the capacitance changes corresponding output signal generated also changes. This capacitance change is marked by the shape of the object and the distance of the object from the sensor.





The sensor consists of an oscillator, trigger circuit, and output switching device. As the target enters the electric field generated by the sensor, the electric field generated gets obstructed by the target resulting in a change in capacitance which results in a change in amplitude of the oscillator circuit. When the target moves further from the sensor the amplitude of oscillation decreases and when the target comes close amplitude increases. This change triggers the trigger circuit resulting in a change in the output of the sensor and thus glowing the output sensor led.

Application

They can be used to detect objects on the conveyor belt.

2. Inductive Proximity Sensor

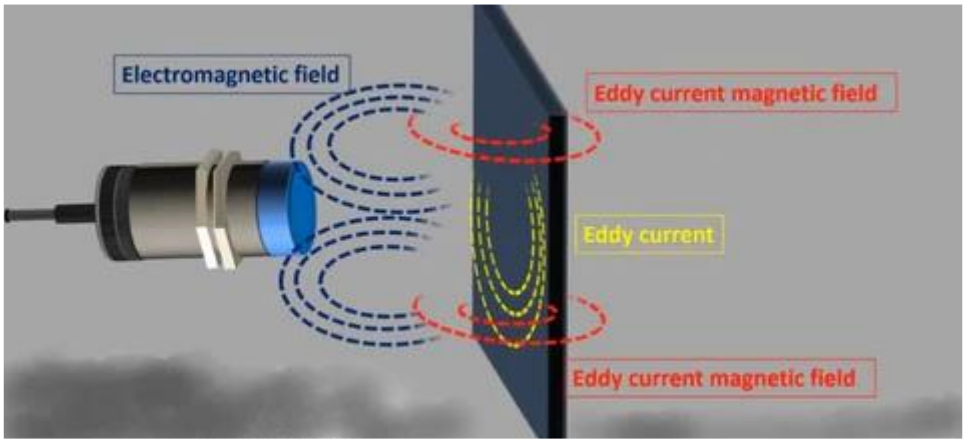
Inductive Proximity Sensor types work on two principles:

Faraday's Law of induction

When an electrical conductor is placed in a magnetic field, eddy currents are produced in the conductor.

Lenz's Law

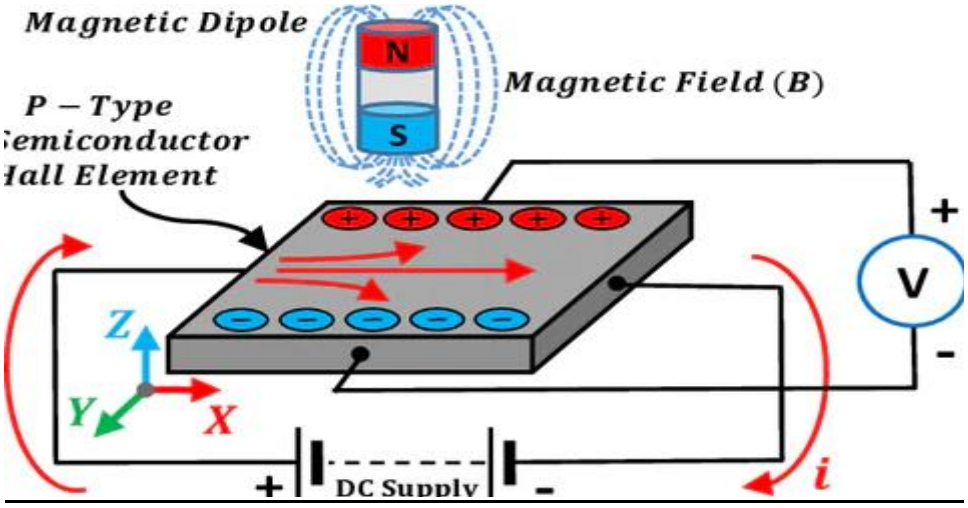
These eddy currents generate a magnetic field that opposes the cause or the magnetic field that generated it.



This type of sensor has a wire wound around typically an iron core. As shown in the above figure, an alternating current is applied to the coil which generates a magnetic field around it. This magnetic field gives rise to an eddy current in the electrical conductor. These eddy currents then generate an eddy current magnetic field that opposes the electromagnetic field from the sensor. When the target is taken close to the sensor the electromagnetic field from the sensor gets reduced, which decreases the amplitude of the electromagnetic field. When it decreases beyond the threshold trigger circuit

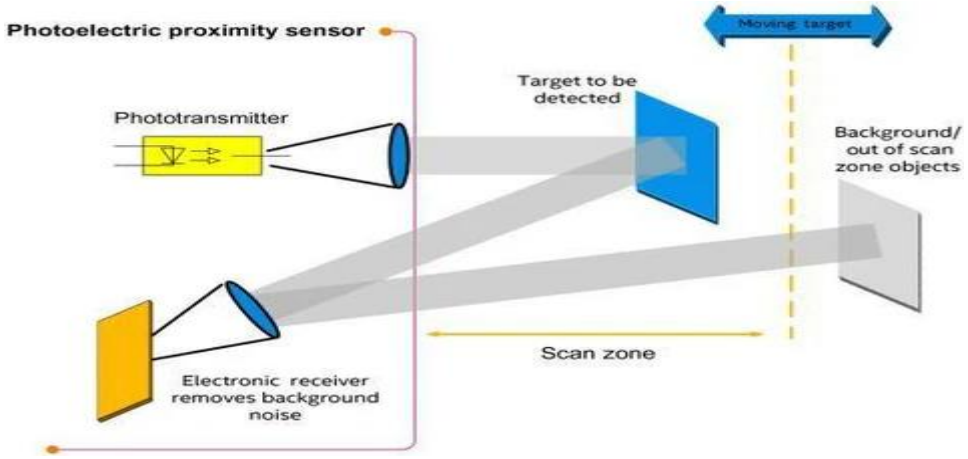
activates the output. This helps to identify that the object has come into the desired vicinity of the sensor. For interfacing proximity sensor with Arduino to get a hands-on experience with this sensor.

3. Magnetic Proximity Sensor



This type of proximity sensor works on the principle of the Hall effect. As shown in the figure, a potential voltage is applied across a hall element such as a hall element made of tin. So positive and negative charges start flowing. Now, when a magnet is brought into the vicinity of the hall element material, negative and positive charges get accumulated across the opposite ends of the plate. When we measure the voltage across it a tiny voltage in microvolts is obtained. This is usually amplified by an amplifier. When the object is taken closer a higher potential gets generated across the plates and when it exceeds the threshold output is set to high. There are various other technologies too used in magnetic proximity sensors such as variable reluctance, reed switches, GMR (Giant Magneto-Resistive effect) inductive.

4. Optical Proximity Sensor



Optical proximity sensor types consist of a transmitter and a receiver. The transmitter circuit consists of a photo transmitter that sends light out by converting electrical energy into light. This light gets bounced back from the target which is received by the photo receiver. There are three types of optical proximity sensors: through-beam sensor, retro-reflective sensor, and diffuser sensor.

5. **Ultrasonic Proximity Sensor**



An ultrasonic proximity sensor consists of a transmitter and receiver. As shown transmitter sends out a sound wave and the receiver detects the reflected wave from the object. You can refer to this guide Arduino Interfacing with an ultrasonic sensor if you want to try something hands-on.

GPS sensor

A GPS sensor is a device that receives signals from GPS satellites to determine a precise location on Earth's surface. This allows the receiver to determine its latitude, longitude, and altitude.

A GPS sensor consists of a surface-mount chip that processes signals from GPS satellites using a small, rectangular antenna, often mounted on the top of the GPS chip.

How it works

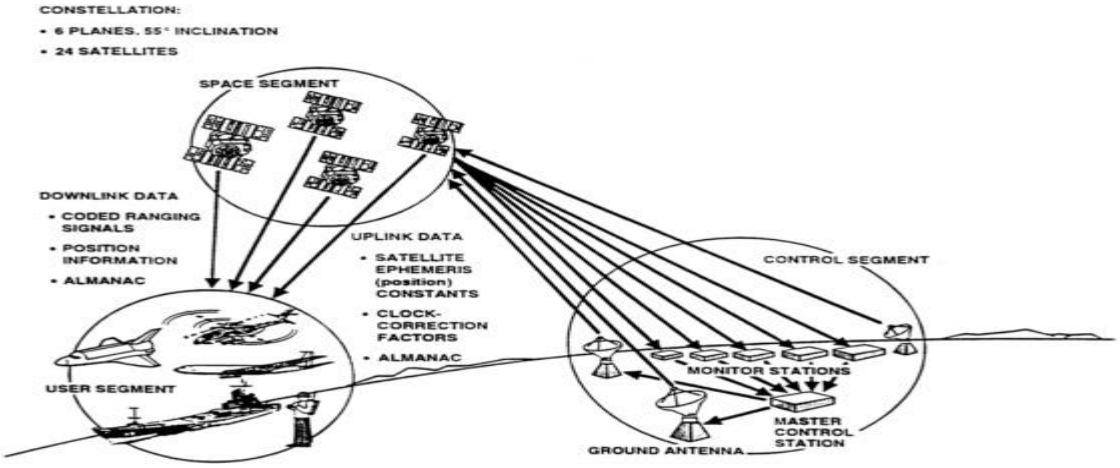
1. **Satellite Signals:** A network of satellites orbit Earth, constantly broadcasting signals containing their orbital position and precise time data.
2. **Receiver Acquisition:** A GPS receiver in a device (like a smartphone or dedicated GPS unit) listens for and receives these signals from at least four satellites simultaneously.
3. **Distance Calculation:** The receiver calculates the distance to each satellite by measuring the time difference between when the signal was sent by the satellite and when it was received by the device. This process uses the speed of the radio signal (approximately the speed of light).
4. **Trilateration:** Using the distances from at least three satellites and their known positions, the receiver uses a mathematical principle called trilateration to pinpoint its own location on Earth's surface. A fourth satellite is often used to improve accuracy and calculate altitude.

Components of a GPS

The GPS system is comprised of three main segments:

- **Space Segment:** This refers to the satellites themselves, which transmit signals to Earth. The space segment contains about 31 satellites as of August 2018, located in orbit about 12,500 miles above the Earth. Hence, each of these satellites circles twice in 24 hours.
- **Control Segment:** This includes ground stations that monitor the satellites, track their orbits, and transmit updated navigation data. The control segment contains command, control, and monitoring stations.

- **User Segment:** This comprises the GPS receivers and transmitters found in devices like smartphones, cars, and dedicated GPS units. The user segment consists of receiving devices (e.g., both government and private).



Applications

GPS sensors are used in various fields:

- **Navigation:** Guiding people and vehicles from one location to another.
- **Tracking:** Monitoring the movement of objects or people, such as in fleet management or personal safety devices.
- **Mapping and Surveying:** Creating and updating maps and for precision surveying.
- **Timing:** Providing highly accurate time signals used in various applications like telecommunications and financial systems.

Accuracy

While GPS technology has become incredibly sophisticated, several factors can affect its accuracy:

- **Environmental Factors:** Tall buildings, dense foliage, mountains, and even weather conditions can obstruct or weaken GPS signals.
- **Satellite Geometry:** The arrangement of satellites in relation to the receiver can also impact accuracy. Optimal satellite geometry provides better accuracy.
- **Signal Quality and Receiver Design:** Strong, clear signals are crucial for accurate readings. High-quality receivers with advanced processing capabilities can process signals more effectively and reduce errors.
- **Signal Interference:** When signals reflect off surfaces like buildings or mountains before reaching the receiver, it can cause multipath interference and errors in location calculations.
- **Advanced Technologies:** Techniques like Differential GPS (DGPS) or Real-Time Kinematic (RTK) GPS use augmentation systems or correction services to improve accuracy, sometimes to centimeter-level precision.

Bluetooth Sensor

Bluetooth sensors are devices equipped with Bluetooth technology to measure and transmit various types of data wirelessly. This eliminates the need for physical cables and enables data collection from remote or difficult-to-access locations. Bluetooth sensors are devices that use Bluetooth technology to transmit data to other devices, such as smartphones, tablets, or computers. They can measure various physical parameters like: Temperature, Humidity, Pressure, Motion, Light.

How they works:

- **Bluetooth Technology:** They utilize Bluetooth Low Energy (BLE), a power-efficient wireless communication protocol, making them suitable for battery-operated applications.
- **Data Transmission:** Sensors transmit measured values to compatible devices like smartphones, gateways, or single-board computers that can process the data, according to getefento.com.
- **Network Connectivity:** Bluetooth sensors can connect via peer-to-peer or mesh networks, allowing for flexible and scalable deployments.

Types of Bluetooth sensors

- **Vibration Sensors/Accelerometers:** These sensors measure vibrations and record data like FFT (Fast Fourier Transform) from multiple axes (X, Y, and Z).
- **Temperature Sensors:** They can be contact-based for surfaces or non-contact (infrared) for remote temperature readings.
- **Humidity Sensors:** Often combined with temperature sensors, they measure environmental humidity.
- **Speed Sensors (RPM):** These use Hall effect switches and magnets to precisely measure rotations per minute.
- **Current Sensors:** They monitor electrical current and power consumption in devices and machinery.
- **General Purpose Sensors:** These can be adapted to monitor various signals and are compatible with other types of sensors like displacement or proximity sensors.
- **Other Types:** You can also find Bluetooth sensors for fuel level, tilt angle, and pulse detection.

Key features and benefits

- **Wireless Connectivity:** Eliminates wiring complexities and allows flexible placement.
- **Low Power Consumption**: BLE's design enables long battery life, with some sensors lasting up to 5 years.
- **Easy Installation:** Their wireless nature makes them straightforward to set up and deploy.
- **Remote Monitoring:** Access sensor data from a distance, which is particularly beneficial in hard-to-reach locations or hazardous environments.
- **Data Collection and Analysis:** Sensors facilitate the collection of various types of data for analysis, enabling condition monitoring and predictive maintenance in industrial settings.

Applications

- **Industrial Monitoring:** Used for online condition monitoring of machinery, including vibration analysis, temperature checks, and current monitoring.
- **Logistics and Transportation:** Track fuel consumption, monitor temperature and humidity in containers, and detect tilt angles in vehicles for freight transportation.
- **Smart Home Automation:** Connecting various devices like door and window sensors, lighting, and heating to a central controller or smartphone.
- **Healthcare:** Monitoring a wide range of body parameters using sensors with Bluetooth connectivity.
- **Retail and Marketing:** Bluetooth beacon sensors provide location-based information to nearby mobile devices, enabling targeted advertisements, customer tracking, and insights into customer behaviour.
- **Environmental Monitoring:** Used to monitor temperature, humidity, and air quality in various settings.

Ultrasonic and Microwave sensor

Both ultrasonic and microwave sensors are used for object detection, distance measurement, and motion sensing, but they employ distinct technologies and excel in different scenarios.

Ultrasonic sensors

Principle: Emit high-frequency sound waves (ultrasound) and measure the time it takes for the waves to reflect off an object and return to the sensor (Time-of-Flight or ToF).

Strengths:

- Accurate distance measurement: Offer precise distance readings, even at close range.
- Not affected by object color or transparency: Can reliably detect objects regardless of their color, transparency, or surface properties (as long as they reflect sound waves).
- Good for harsh environments: Less affected by dust, smoke, mist, and dirt compared to optical sensors.
- Versatility: Suitable for detecting solid, liquid, granular, or powder materials.

Limitations:

- Limited range: Generally designed for short- to medium-range detection, typically up to 10 meters, with some specialized models reaching 20 meters.
- Sensitivity to soft or angled surfaces: Soft or porous materials, like cloth or foam, can absorb sound waves, reducing detection accuracy. Angled surfaces can deflect the waves away from the sensor.

- Environmental influences: Temperature and humidity can affect the speed of sound and thus impact measurement accuracy, though some sensors offer temperature compensation.
- Cross-talk interference: Multiple ultrasonic sensors in close proximity can interfere with each other's signals.
- Blind spot: A small area directly in front of the sensor where it cannot detect objects due to the time required for signal transmission and reception.

Applications:

- Robotics: Obstacle avoidance and navigation.
- Industrial automation: Object detection, counting, positioning, and level measurement in tanks and silos.
- Automotive: Parking assistance systems and blind spot monitoring.

Microwave sensors

Principle: Emit microwave signals and measure the reflection off moving objects, relying on the Doppler effect (change in frequency of a wave due to motion) to detect movement.

Strengths:

- Wide detection range: Can detect objects over a larger area compared to ultrasonic sensors, sometimes up to 16 meters or more, depending on the model and environment.
- Penetration capabilities: Can detect motion through non-metallic obstacles like walls or glass, according to Sensinova India.
- High sensitivity: Can detect even subtle movements.
- Works in various environmental conditions: Less affected by temperature, humidity, light, or dust than some other sensors.

Limitations:

- Prone to false alarms: High sensitivity can lead to false triggers from minor movements like a swaying tree branch or animals.
- Higher power consumption: Generally consume more power than ultrasonic sensors.
- Limited penetration of metal: Microwaves cannot penetrate metal objects, so detection behind metal obstructions is not possible.
- Difficulty in defining range: Microwave beams do not have sharply defined edges, making precise range definition challenging.

Applications:

- Security systems: Motion detectors, alarms, and access control.

- Traffic monitoring: Vehicle speed, traffic flow, and occupancy detection.
- Industrial automation: Object detection, positioning, and movement monitoring.
- Automated lighting: Turning lights on and off based on occupancy.

When to choose which sensor

- **Ultrasonic sensors:** Ideal for applications requiring accurate short- to medium-range distance measurements, particularly in situations where object colour or transparency might be an issue. They are well-suited for industrial automation, robotics, and parking assistance systems where they need to operate in dusty or low-light conditions.
- **Microwave sensors:** Preferable for applications needing wide-area coverage and the ability to detect movement through non-metallic obstructions. They are effective in security systems, traffic monitoring, and automated lighting where higher sensitivity is advantageous, according to Integral LED.

LIDAR Sensor

LiDAR, which stands for Light Detection and Ranging, is a remote sensing technology that utilizes laser light to measure distances and construct 3D models of objects and environments. It works by emitting laser pulses and then measuring the time it takes for these pulses to reflect off objects and return to the sensor. This "time-of-flight" data, combined with information about the sensor's position and orientation (often using GPS and IMUs), allows for the creation of detailed 3D representations called point clouds.

Key principles

- **Active sensing:** Unlike cameras that passively capture existing light, LiDAR actively emits its own light source (laser pulses), allowing it to function in various lighting conditions, including darkness.
- **Time-of-flight measurement:** The core principle involves precisely measuring the time it takes for a laser pulse to travel to an object and return, enabling accurate distance calculations.
- **Point clouds:** The collected distance and positional data are compiled into a dense collection of 3D points, forming a point cloud that visually represents the scanned environment.

Types of LiDAR

LiDAR sensors are broadly classified based on their platform:

- **Airborne LiDAR:** These sensors are mounted on aircraft (like planes or helicopters) and are used for surveying and mapping large geographical areas.
    - Topographic: Uses near-infrared lasers to map land and features above ground.
    - Bathymetric: Uses water-penetrating green lasers to map underwater environments like coastlines and riverbeds.
- **Terrestrial LiDAR:** These are ground-based systems, either stationary or mounted on vehicles.

- Static: Fixed in one location for detailed and repeated scans of smaller areas, suitable for archeological sites, construction monitoring, or assessing geological hazards.
- Mobile: Mounted on moving vehicles (e.g., cars, robots) for real-time data collection in applications like autonomous driving, robotics navigation, and transport planning.

Applications:

LiDAR technology finds applications across a vast range of industries and fields:

- **Autonomous vehicles and robotics:** For perception, navigation, obstacle detection, and mapping in self-driving cars, industrial robots, and UAVs.
- **Mapping and surveying:** Creating high-resolution 3D maps, digital elevation models (DEMs), and digital terrain models (DTMs) for urban planning, infrastructure development, and environmental monitoring.
- **Forestry and agriculture:** Assessing forest health and structure, monitoring crop growth, and optimizing farming practices.
- **Archaeology:** Discovering hidden archaeological sites and structures beneath dense vegetation.
- **Environmental research:** Monitoring glacier melt, sea-level changes, and air quality.
- **Weather forecasting:** Measuring atmospheric parameters like cloud height, wind speed, and air density.
- **Other applications:** Flood forecasting, law enforcement (speed guns, crime scene investigation), defense and military (reconnaissance, target identification), mining, space exploration (planetary mapping, spacecraft navigation), and more.

Advantages:

LiDAR offers several benefits:

- **High accuracy and precision:** Provides detailed and reliable measurements.
- **Speed:** Quickly captures data over large areas.
- **3D capabilities:** Creates comprehensive 3D models.
- **Operates in various lighting conditions:** Effective day or night, and in certain weather conditions.
- **Penetrates vegetation:** Can generate ground elevation data even through dense forest canopies.

Disadvantages:

However, LiDAR also has limitations:

- **Cost:** LiDAR equipment can be expensive.
- **Weather sensitivity:** Performance can degrade in severe weather conditions like heavy rain, snow, or fog due to signal absorption and scattering.

- **Data processing complexity:** Handling and analysing large point cloud datasets requires specialized software and expertise.
- **Limited object recognition (compared to cameras):** While providing excellent 3D structure, LiDAR may struggle to distinguish between certain objects or read signs like traffic lights without the aid of cameras.

Optical and Radiation Sensor

Optical and radiation sensors are distinct but sometimes overlapping fields in sensing technology. Optical sensors detect and measure light, which is a form of electromagnetic radiation. Radiation sensors, however, are broader and encompass detection of various types of electromagnetic radiation, including X-rays, gamma rays, and ultraviolet radiation, as well as particle radiation like alpha and beta particles.

I. Optical sensors

Optical sensors are electronic detectors that convert light or changes in light into electrical signals. They are generally part of a larger system that includes a light source, the optical sensor itself, and a measuring device.

A. Working principles

Optical sensors rely on the interaction of light with the target object, utilizing optical properties like reflection, absorption, or transmission. The sensor illuminates the target with light (e.g., from an LED or laser) and analyses the resulting optical signals to determine characteristics of the target object, such as its presence, distance, position, colour, or other properties.

Common light sources include LEDs and lasers. Photodetectors, such as photodiodes or phototransistors, capture the variations in light intensity or other properties and convert them into measurable electrical signals.

B. Types of optical sensors

- **Through-Beam Sensors:** Emitter and receiver are placed opposite each other, and an object is detected when it interrupts the light beam.
- **Diffuse Reflective Sensors:** Emitter and receiver are integrated into a single unit and detect light reflected directly off the target.
- **Retroreflective Sensors:** Emitter and receiver are housed together, and the sensor uses a reflector opposite to detect objects that block the reflected light beam.
- **Fiber-Optic Sensors:** Utilize optical fibers to transmit and receive light, suitable for detection in tight, harsh, or high-temperature environments.
- **Laser Sensors:** Employ focused laser beams for high-precision detection and distance measurement.
- **Proximity Optical Sensors:** Detect nearby objects by measuring reflected or interrupted light, commonly used in robotics and automation.

- **Optical Encoders:** Measure rotation, position, speed, or direction using a patterned disc, light source, and sensor.

C. Applications of optical sensors

Optical sensors are ubiquitous in various applications, including:

- **Industrial Automation:** Product positioning, item counting, and quality control on manufacturing lines.
- **Consumer Electronics:** Facial recognition, automatic lighting adjustment, and gesture control in smartphones.
- **Medical Devices:** Pulse oximeters for measuring blood oxygen levels, breath analysis, and heart rate monitoring.
- **Environmental Monitoring:** Measuring light intensity or water clarity.
- **Automotive Safety:** Lane detection, parking assistance, and collision prevention systems.
- **Security Systems:** Motion detection and surveillance.

II. Radiation sensors

Radiation sensors, also known as radiation detectors, are devices used to detect and measure ionizing radiation, which includes electromagnetic waves like X-rays and gamma rays, and particle radiation like alpha and beta particles.

A. Working principles

Radiation sensors typically operate by detecting the interaction of radiation with a specific material within the detector, causing a measurable signal. This interaction can involve ionization (where radiation dislodges electrons from atoms) or excitation (where radiation causes atoms to move to higher energy states).

B. Types of radiation sensors

- **Gas-Filled Detectors:** Contain a gas (e.g., argon or helium) that gets ionized when radiation passes through, and the amount of ionization is measured. A well-known example is the Geiger-Müller counter.
- **Scintillation Detectors:** Contain a scintillating material (like a sodium iodide crystal) that emits light flashes when struck by radiation particles. A photomultiplier tube then converts these light flashes into an electrical signal.
- **Solid State Detectors:** Employ semiconductor materials (e.g., silicon or germanium) to detect radiation by the generation of electron-hole pairs as radiation passes through a reverse-biased p-n junction diode, leading to a current pulse that can be measured.

C. Applications of radiation sensors

Radiation sensors are essential for monitoring radiation levels in various applications:

- **Industrial Radiation Monitoring:** Monitoring at nuclear power plants, waste storage facilities, and in food irradiation processes.

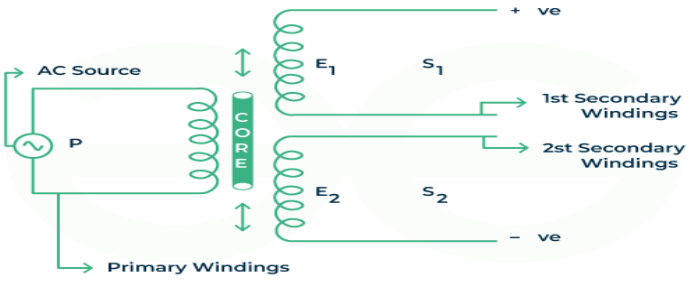
- Space Exploration: Monitoring cosmic rays in spacecraft to protect astronauts and sensitive equipment.
- **Medical Radiation Dosimetry:** Measuring radiation doses in radiotherapy to ensure accurate treatment delivery and patient safety.
- **Environmental Monitoring:** Assessing baseline radioactivity data in areas near industrial facilities or monitoring for potential environmental contamination.
- **Security:** Detecting radioactive materials and potential threats.

III. Key differences between optical and radiation sensors

- **Target:** Optical sensors focus on detecting light, while radiation sensors detect various forms of radiation, including visible light, but also X-rays, gamma rays, and particle radiation.
- **Mechanism:** Optical sensors typically use the transmission and reception of light and variations in optical properties, while radiation sensors often rely on ionization or excitation of materials.
- **Penetration:** Different types of radiation have varying penetration capabilities, which dictates the type of sensor used and the required shielding. Optical sensors are primarily concerned with the interaction of light within a defined medium.

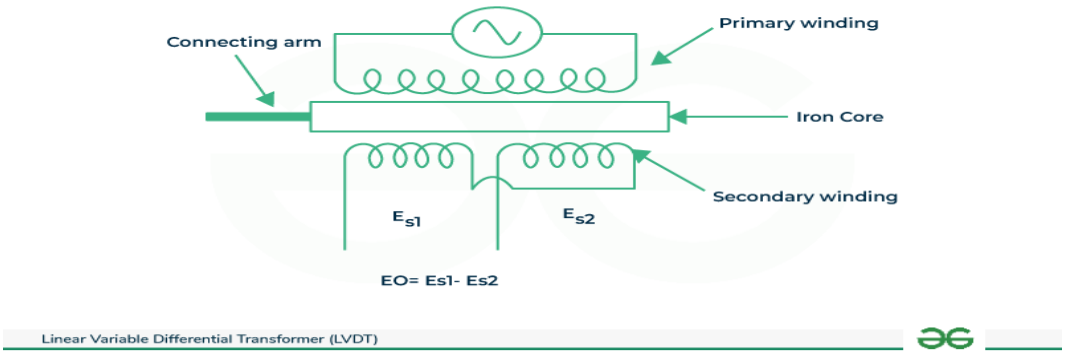
Linear Variable Differential Transformer (LVDT)

LVDT stands for Linear variable differential transformer. It is one of the major types of Inductive transducers. LVDTs are considered the most accurate inductive sensors that measure displacement according to the polarity and magnitude of the net induced electromotive force (EMF) and are therefore also known as linear variable displacement sensors.



Linear Variable Differential Transformer (LVDT)

Construction of LVDT



The transformer and LVDT share a similar construction. It consists of one primary winding(P) and two secondary windings (S1 & S2). The primary and secondary windings are bounded by a hollow cylinder, known as the former. The primary winding is at the center and the secondary windings are present on both sides of the primary winding at an equal distance from the center. Both the secondary windings have an equal no. of terms and they are linked with each other in series opposition, i.e. they are wounded in opposite directions, but are connected in series with each other.

The entire coil assembly remains stationary during distance measurement. The moving part of the LVDT is an arm made of magnetic material.

Working Principle of LVDT

The working of LVDT is based on Faraday's law of electromagnetic induction, which states that "the electrical power in the network induction circuit is proportional to the rate of change of magnetic flux in the circuit."

As the primary winding of LVDT is connected to the AC power supply, The alternating magnetic field is produced in the primary winding, which results in the induced EMF of secondary windings.

Let's assume that the induced voltages in the secondary windings S1 & S2 are E1 & E2 respectively. Now according tousing the rate of change of magnetic flux i.e. dΦ/dt is directly proportional to the magnitude of induced EMF i.e E1 and E2.

The total output voltage Eo in the circuit is given by Eo = E1-E2

Depending on the position of the core some cases arise:

Case 1: When The Core is Moving Towards S1

When the core of LVDT moves towards the second winding S1 then the flux linkage S will be more as compared to S2. The EMF induced in S1 will be more than the EMF of S2. Hence E1 is greater than E & net differential voltage Eo(E1-E2) will be +ve. The means output voltage Vo will be in phase with input AC voltage.

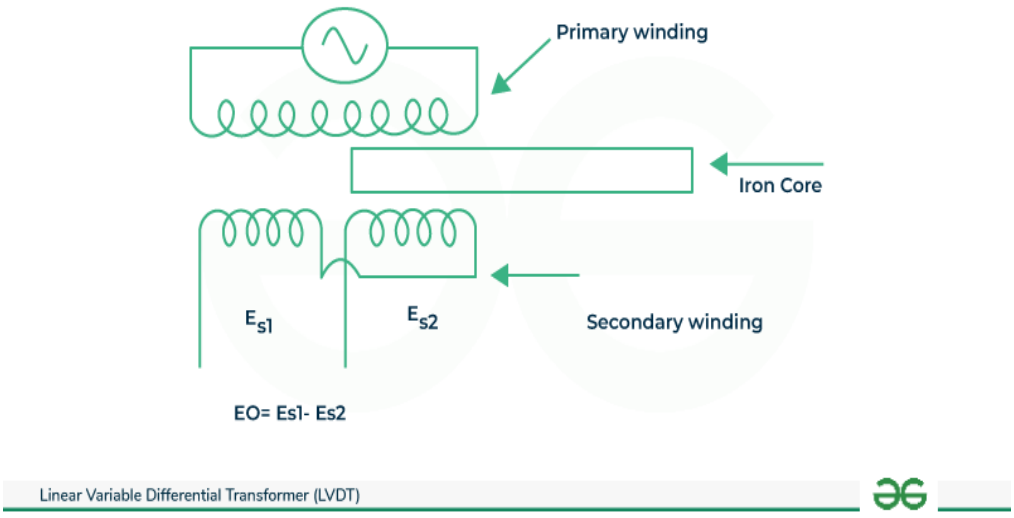
Case 2 : When the core is positioned at its null position

When the core is at a null position then the flux generated in both the secondary windings will be the same. The induced EMF E1 & E2, and both the windings will be the same. Hence the net differential output voltage Eo will be 0. It shows 0 displacement of the core.

Case 3 : When The Core Moving Towards S2

When the core of LVDT moves towards secondary winding S2 then the flux linkage with S2 will be more than S1. It means the EMF induced in S2 will be more than the induced EMF of S1

Hence E2 is greater than E1 & net differential voltage Eo (E1-E2) will be negative. It means the output voltage will be out of phase input AC voltage.



Output of LVDT

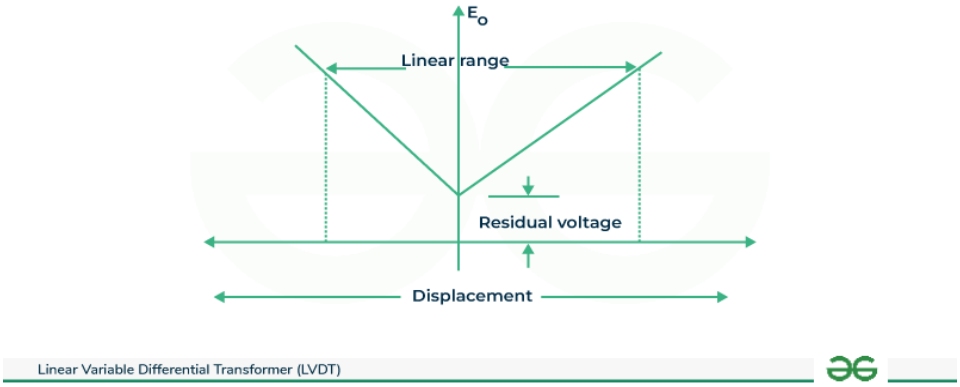
- The output of a Linear Variable Differential Transformer (LVDT) is an AC voltage that is proportional to the displacement or position of its core.
- A zero-differential output voltage is produced when the core is in the center, or null position, where the induced voltages in the two secondary coils are equal. The induced voltages in the secondary coils become unequal as the core moves away from the null position, and the differential output voltage increases proportionately.

So in a nutshell we can conclude that the output of an LVDT is an AC voltage and the magnitude and other measurements of this output voltage provide insightful information about the direction and amount of displacement which is later inspected and fixed if any problem is detected.

Uses of LVDT

Linear Variable Differential Transformers (LVDT) are mainly used for work or motion or movement in many industries and research, they are also used in medicine and patient monitoring similarly in many applications because of its ability of unique features and advantages. It offers high precision and accuracy as it has non-coated sensors it can also vary with measurement ranges like the miniature or subminiature LVDTs are suitable for applications with extremely limited space or where fine-scale displacement measurements are necessary. its versatile nature, Long term stability, and reliability are what make it useful and make it a preferred choice for displacement and position sensing in numerous applications.

Characteristics of LVDT



Display the displacement versus output voltage graph for both left and right movement and how the output voltage varies with linear displacement.

The LVDT is an electromechanical device that generates an AC voltage output in direct proportion to the ferromagnetic core and relative displacement of the transformer. The ability of LVDT sensors to function in harsh environments with high levels of vibration and shock is a crucial feature.

LVDT Specifications

As we can understand the Linear Variable Differential Transformer's (LVDT) specifications can change based on the model and the requirements of the application. some common specifications of the LVDTs are:

- Linearity
- Sensitivity
- Resolution
- Electrical Output
- Temperature Range etc.

Advantages of LVDT

- **High output:** For minute variations in the magnetic core position, LVDTs provide a high output.
- **Low hysteresis:** LVDTs are highly repeatable due to their extremely low hysteresis.
- **Low electrical noise:** Because LVDTs have sensing coils with low impedance, they can produce extremely low electrical noise levels.
- **Less power Consumption:** LVDT's consume less power as compared to other Transducer's

Disadvantages of LVDT

- Since LVDT is an inductive transducer it is sensitive to the stray magnetic field, hence an extra setup is required to protect from stray magnetic field.
- As LVDT is an electromagnetic device, it is also affected by vibrations and temperature.

Applications of LVDT

- It is mostly used in industries in the field of Automation, Aircraft, Turbines, Satellite, Hydraulics etc.
- LVDT is used to measure physical quantity such as force, tension, pressure weight, etc. here LVDT is used as a secondary transducer.
- LVDT plays important role in geotechnical Instrumentations, as it is used for Monitoring Ground Movements, Landslides and Structural Stability
- LVDT plays an important role in the marine and offshore industry by Monitoring the Movements and Positions of ships and Underwater Structures.
- LVDT Plays an important role in Power Generation as it monitors the Critical Components in turbines and generators.

Rotary Variable Differential Transformer (RVDT)

The transformer which senses the angular displacement of the conductor is known as the Rotary Variable Differential Transformer or RVDT.

It is the type of electromechanical transducer which gives the linear output proportional to the input angular displacement.

The circuit of RVDT is shown in the figure below. The working of the RVDT is similar to the LVDT. The only difference is that the LVDT uses the soft iron core for measuring the displacement, whereas the RVDT uses the cam shape core rotated between the primary and secondary winding with the help of the shaft.

Theory of RVDT

The E_{S1} and the E_{S2} are the secondary voltage, and it varies with the angular displacement of the shaft.

$$\theta = G.\left(\frac{E_{S1} - E_{S2}}{E_{S1} + E_{S2}}\right)$$

The G is the sensitivity of the RVDT. The secondary voltage is determined by the help of equation shown below.

$$E_{S2} = E_{S1} \pm G.\theta$$

The difference between $E_{S1} - E_{S2}$ gives a proportional voltage.

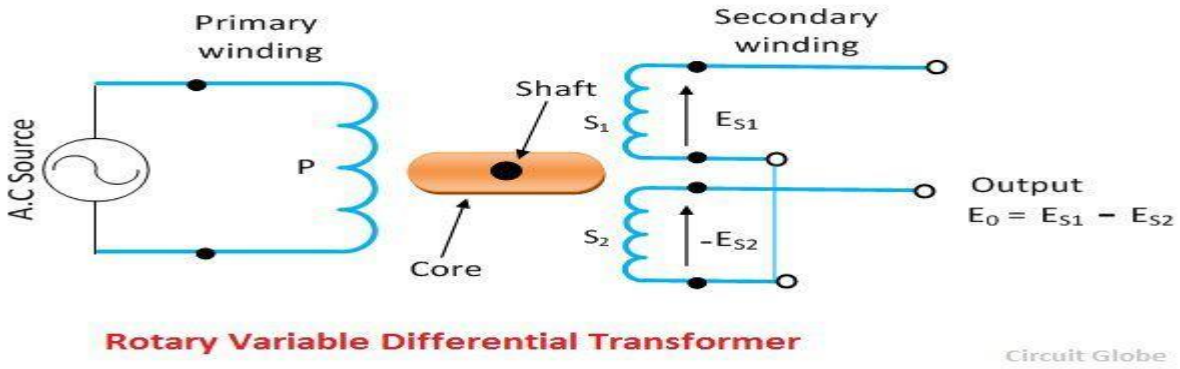
$$\Delta E_s = 2.G.\theta$$

The sum of the voltage is given by constant C.

$$C = \sum E_s = 2.E_{S0}$$

Operation of LVDT

When the core is in the null position, the output voltages of the secondary winding S_1 and S_2 are equal and opposite. The net output of null position is zero. Any angular displacement from the null position will give the differential output voltage. The angular displacement is directly proportional to the differential output voltage. The response of the RVDT is linear.



The differential output voltage of the transformer increases when the shaft rotates in a clockwise direction. And it decreases when the shaft moves in an anti-clockwise direction. The magnitude of the output voltage depends on the angular displacement and the direction of the shaft.

The typical RVDTs are linear over a +40 or -40 degrees, Sensitivity is about 2mV to 3mV per degree of rotation and the input voltage range is 3V RMS at frequency ranges from 400Hz to 20kHz. Based on the movement of the shaft in the transformer, the three conditions will be produced such as

- When the Core is at Null Position
- When the Core Rotates in Clockwise Direction
- When the Core Rotates in Anticlockwise Direction

When the Core is at Null Position

In the first condition, when the shaft is placed at the null position then the induced e.m.f in the secondary windings are similar although reverse in phase. Thus, the differential o/p potential will be zero, and the condition will be E1 = E2, where E0 = E1-E2 =0

When the Core Rotates in Clockwise Direction

In the second condition, when the shaft rotates in the direction of clockwise; more section of the core will enter across the primary winding. Therefore, the induced e.m.f across the primary winding is higher than secondary winding. Hence, the differential o/p potential is positive, and the condition will be E1 > E2, where E0 = E1-E2 = positive.

When the Core Rotates in Anticlockwise Direction

In the third condition, when the shaft rotates in the direction of anticlockwise, more section of the core will be entered across the secondary winding. Thus, the induced e.m.f across the secondary coil is higher than the primary coil. Hence, the differential o/p potential is negative that means 1800 phase shift, and the condition will be E1 < E2, where E0 = E1-E2 = negative.

RVDT Advantages and Disadvantages

The advantages of RVDT include the following.

- The consistency of RVDT is high
- The exactness of RVDT is high
- The lifespan is long
- The performance is repeatable
- The construction is compact and strong
- Durability
- Low cost
- Easy to handle electronic components
- Resolution is infinite
- Linearity is Excellent
- A wide range of dimension ranges

The disadvantages of RVDT mainly include the following

- The contact among the measuring exterior as well as the nozzle is not possible for all time.
- The output of the RVDT is linear (about +40 or -40 degrees), so it restricts the usability.

RVDT Applications

The applications of RVDT include the following.

- Fuel Valves as well as Hydraulic
- Modern machine tools
- Controls Cockpit
- Controls Fuel
- Brake with cable systems
- Engines bleed air-systems
- Robotics
- Aircraft and Avionics
- Process Control industry
- Weapon and Torpedo Systems

- Engine fuel control
- Nose wheel steering systems
- Fly by wire systems
- Push reverser
- Actuators for controlling Flight as well as Engine
- Ecological control systems